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In union lies strength: Collaborative competence in new product development and its performance effects

Anant A. Mishra [1], Rachna Shah *

Carlson School of Management, University of Minnesota, 321 Nineteenth Avenue South, Minneapolis, MN 55455-0438, United States

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ABSTRACT

It is widely recognized that new product development (NPD) is a highly interdependent process, yet efforts to empirically model the interdependence and examine its effect on firm performance are scarce. Our study addresses this research gap. We model firms' abilities to collectively collaborate with suppliers, customers, and internal employee teams in NPD as *collaborative competence* and examine its impact on project and market performance. Using responses collected from 189 NPD managers, we find empirical evidence for collaborative competence and its differential impact on project and market performance. Specifically, we find that collaborative competence has a direct impact on project performance, but its impact on market performance is indirect, mediated through project performance. The results have significant managerial implications; achieving superior market performance from inter- and intra-organizational involvement is contingent on achieving superior project performance, and companies that fail to achieve desired project performance outcomes will also fail in achieving market performance goals.

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1. Introduction

New product development (NPD) is a key source of competitive advantage for individual firms (Verona, 1999). Successful NPD requires firms to develop routines and practices to collaborate with suppliers, customers, and internal cross-functional employee teams. While some firms are able to involve key suppliers in their NPD endeavors, others are more effective in collaborating with customers. Still others develop expertise in leveraging advantages garnered from involving their own employees in product development teams. Few firms, however, develop the competence to engage suppliers, customers, and internal employees simultaneously in their NPD projects. We believe that the ability to simultaneously collaborate with suppliers, customers, and cross-functional teams in NPD is a valuable – yet rare – firm level capability. We label it *collaborative competence* in this study.

The notion that focal firms need to integrate externally with their suppliers and customers and internally across different functions to achieve NPD success is not a new idea; it has a rich history in the organizational strategy (Bajaj et al., 2004; Grant, 1996; Bensaou and Venkatraman, 1995; Kogut and Zander, 1992) and supply chain literature (Frohlich and Westbrook, 2001; Hammel and Kopczak, 1993). Frohlich and Westbrook (2001) demonstrate empirically that firms with the widest arc of integration (i.e., firms that integrate both suppliers and customers into the activities of the focal firm) have the strongest association with performance improvement as compared to firms that integrate only suppliers or only customers. The practitioner literature also highlights the superior performance benefits of simultaneously involving multiple

* Corresponding author. Tel.: +1 612 624 4432; fax: +1 612 624 8804.
E-mail addresses: mish0049@umn.edu (A.A. Mishra), shahx024@umn.edu (R. Shah).
[1] Tel.: +1 612 626 9761; fax: +1 612 624 8804.

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stakeholders in the NPD process (e.g., Carlile, 2004; Orlikowski, 2002). Despite the increased attention to collaboration in academic and practitioner literature, there is relatively little rigorous empirical research identifying the managerial actions and mechanisms that underlie collaborative competence.

In this study, we define collaborative competence as the ability to simultaneously involve key stakeholders in the NPD process and examine its effect on performance. We begin by developing a conceptual foundation for the underlying dimensional structure of collaborative competence by identifying three sets of collaborative practices related to suppliers, customers, and internal cross-functional employee teams. Following this, we empirically examine whether the three sets of collaborative practices indeed constitute collaborative competence in an NPD context. To quantify the performance benefits associated with collaborative competence, we operationalize performance as *project performance* and *market performance*. We investigate direct effects of collaborative competence on project and market performance and also test for the mediating role of project performance between collaborative competence and market performance. Complementarity theory and resource-based view (RBV) provide the theoretical grounding for our investigation.

We use project level data collected from 189 manufacturing plants in three industries and six countries to test our conceptual framework linking collaborative competence with project and market performance. Our results make three contributions to the literature. First, by empirically modeling the interdependencies among collaborative practices, we accurately replicate the highly interdependent nature of the product development process. Our second contribution lies in demonstrating that collaborative competence has a different impact on project and market performance measures; our results indicate that collaborative competence has a strong direct effect on project performance and an indirect effect on market performance, mediated by project performance. A third contribution of our research is related to the generalizability of empirical results obtained from using a multi-industry, multi-country sample. As a set, our results extend our understanding of the role of collaboration in NPD process and its specific links with performance.

The rest of the paper is organized in the following manner. In Sections 2 and 3, we review the existing NPD literature related to collaborative practices and present three research hypotheses. In Section 4, we describe the research design and measures. Section 5 outlines the empirical methods we use to evaluate the hypotheses and results followed by Section 6, in which we discuss our findings and their implications for research and practice. Finally, limitations and future research ideas are outlined in Section 7.

2. Collaboration in NPD

The literature on collaboration in NPD has a rich history with the information processing theory (IPT) (Galbraith, 1977; Tushman and Nadler, 1978; Daft and Lengel, 1986) serving as the primary theoretical lens through which collaboration between a firm and its internal and external partners in NPD has been studied. According to this perspective, an organizational design that is efficient in both acquiring and processing rich information is best suited for uncertain, equivocal tasks. Uncertainty arises from "the difference between the amount of information required to perform the task and the amount of information already possessed by the organization" (Galbraith, 1977). Equivocality, on the other hand, arises from the existence of multiple conflicting interpretations about organizational situations (Weick, 1979).

As NPD environments are typically characterized by high levels of both uncertainty and equivocality, an "integrated problem solving" approach that requires the early involvement of key stakeholders in the product development process and permits sharing of critical information upstream and downstream in the product development supply chain as necessary for executing NPD work (Wheelwright and Clark, 1992). Swink (2006, p. 1) states that an organization's ability to *collaborate* is key to its innovative success. It also suggests that collaborative NPD *integrates* many sources of design ideas and data. Building on Swink's conceptualization, we study collaboration in NPD across three interfaces: collaboration between the core design team and the suppliers, customers, and cross-functional teams.

To do so, we conducted an extensive review of organization theory, strategy, and operations management literatures and found that, while the term "collaboration" is widely used in the NPD context, it is frequently used interchangeably with integration and coordination and, to a lesser extent, with cooperation and communication (Gulati et al., 2005; Barki and Pinsoneeault, 2005; Frohlich and Westbrook, 2001; Van de Ven et al., 1976). Upon reviewing the long and rich evolutionary history associated with this body of literature, we came to the conclusion that it is very difficult, if not impossible, to resolve the differences among these different terms. Moreover, resolving these differences is not central to our research objectives. Rather, given that collaboration across each of the three interfaces has received widespread attention in existing literature, we summarize findings from key studies in Table 1.

With regard to supplier involvement, Clark and Fujimoto (1991), Eisenhardt and Tabrizi (1995), Smith and Reinertsen (1998), Gupta and Souder (1998), all find that involving suppliers in the NPD process is critical to accelerating the pace of product development. Suppliers are more likely to identify potential problems such as contradictory specifications or unrealistic designs, early in the design process. Involving suppliers in the NPD process also opens up outsourcing and external acquisition possibilities, thereby reducing the internal complexity of projects and providing extra personnel to shorten the critical path for NPD projects. Besides shortening development cycles, supplier involvement has been shown to have a positive effect on other measures of performance such as lower development costs (e.g., Kessler, 2000, McGinnis and Vallopra, 1999), improved design-for-manufacturability (e.g., Wasti and Liker, 1997, Swink, 1999) and enhanced product quality (e.g., McGinnis and Vallopra, 1999).

Table 1
Overview of selected literature on collaborative practices[a].

	SI	CI	CFI	Independent variable (s)	Dependent variable(s)	Result[b]
Ancona and Caldwell (1992)			X	An entropy-based diversity index to calculate functional diversity.	Composite measures of performance that captures efficiency, quality, technical innovation, adherence to schedules, adherence to budget, and work excellence.	+ve support
Eisenhardt and Tabrizi (1995)	X		X	No. of stages in which suppliers participated; total no. of functions in every stage of product development.	Development time adjusted by industry.	+ve support only in computer industry sample
Ettlie (1995)			X	Personnel integration that captures the extent of job rotation between design and manufacturing engineering.	Organizational performance measured as sales/employee; development cycle time adjusted by product life cycle.	+ve support
Hartley et al. (1997)	X			Length of supplier involvement.	Perceived contribution to new product design.	Supported
Ittner and Larcker (1997)	X	X	X	Extent of supplier involvement, customer involvement, cross-functional teams in product design.	Organizational performance—composite measure of ROA, sales growth, return on sales, and perceived overall performance.	+ve support only for cross-functional teams
Moffat (1998)		X		Customer input index capturing whether (a) customers were on the team, (b) marketing function was present in the team, (c) customer input was used, and (d) QFD techniques were used.	Project task performance—a composite measure of (a) timeliness of delivery, (b) product quality, (c) design iteration/engineering change order reduction, and (d) internal external customer satisfaction.	Not supported
Swink (1999)	X		X	Extent of supplier influence (single item); extent of manufacturing involvement (three item scale).	New product manufacturability—composite measure of product cost, product quality, manufacturability, etc.	+ve support
Ragatz et al. (2002)	X			Extent of supplier involvement in decision making.	Project outcomes—composite measure of overall satisfaction and degree of goal achievement.	+ve support
Bajaj et al. (2004)		X		Degree of customer interaction measured as the ratio of number of customer signoffs in the design phase to total design budget.	Extent of savings in design time and design cost with respect to budgeted goals.	+ve support when specialization and oversight is high
Koufteros et al. (2005)	X	X		Extent of customer integration in NPD efforts, supplier integration in product engineering, and supplier integration in development process.	Product innovation performance; product quality performance	+ve support only for effects of customer integration
Langerak and Hultink (2005)	X	X	X	Extent of (a) supplier involvement, (b) lead user involvement, and (c) stimulating inter-functional cooperation in NPD efforts.	New product (a) development speed—a composite measure of two items that captures the ability to minimize the time in developing new products; (b) profitability—a composite measure of two item that captures the extent to which new product met the profit and margin goals.	+ve support for the effect of supplier involvement on development speed and lead user involvement on both metrics
Petersen et al. (2005)	X			Supplier effort in (a) technical assessment and (b) business assessment.	Project team effectiveness—a composite measure that captures the extent to which the project team functions smoothly, achieves consensus, and makes faster decisions.	+ve support for supplier effort in technical assessment on project team effectiveness

[a] SI: supplier involvement, CI: customer involvement, and CFI: cross-functional involvement.
[b] +ve: positive, -ve: negative.

Similarly, with regard to customer involvement, evidence from case studies indicates that an incomplete and inaccurate assessment of customers' needs can result in extremely high costs at the design stage (von Hippel, 1998; Walker and Tersini, 1992). Involving the customer in the product development process keeps the design team members updated on changing customer tastes and requirements and reduces the high degree of uncertainty accompanying new applications in the industry (MacCormack et al., 2001; Griffin and Hauser, 1993). At the same time, frequent and regular communication with customers prevents what Thomke and Nimgade (2000) refer to as "creeping elegance", or the designers' quest for perfection, which could lead to cost and time overruns. Customer involvement is clearly a critical aspect of the NPD process.

Finally, involving cross-functional teams (i.e., teams comprised of members from engineering, manufacturing, marketing, etc.) in the NPD process is considered beneficial. Increasing functional diversity in cross-functional teams increases the amount and variety of information available to design products, reducing much of the uncertainty and equivocality in the NPD process on one hand and helping project team members understand the design process more quickly and fully from a variety of perspectives on the other (Henke et al., 1993). It also helps identify possible downstream problems such as manufacturing difficulties or potential mismatches with customer requirements (Ancona and Caldwell, 1992; Eisenhardt and Tabrizi, 1995).

In addition to these advantages, researchers have listed potential problems with collaborative initiatives. Hansen and Nohria (2004, p. 11), for instance, theorize that collaboration can be overdone easily and state that employees, in response to collaborative initiatives, may begin to participate in extraneous meetings in which nothing of substance is accomplished. This suggests that collaborating within and across firm boundaries is time-consuming and resource intensive. A focal firm has to commit substantial resources to overcome dissimilar management styles of collaborating partners for collaboration to succeed. Other problems include infringement of intellectual property rights for participating firms once the collaboration period is over and behavioral misjudgments that can occur during team interactions. Despite these disadvantages, the benefits from collaboration far outweigh its potential costs.

The overarching consensus gleaned from the NPD literature is that collaboration with each of the three interfaces provides both operational and financial/market performance benefits. Our review also supports Gerwin and Barrowman's (2002) assertion that while researchers recognize the importance of these sets of practices and their interdependencies, they examine them in isolation. Such isolated study leads to potentially incorrect conclusions. Gerwin and Barrowman state that most empirical studies tend to disregard the "interpretive aspect of coordination" (p. 940), which involves the need to reconcile the differing meanings across the three interfaces for fairly well-agreed upon goals and tasks. We attempt to fill this gap by examining them together in one study and seek to extend this literature stream by examining the performance implications arising from simultaneously pursuing collaborative NPD practices across the three interfaces.

3. Hypothesis development

We use RBV and complementarity theory to develop our conceptual framework. Researchers invoking an RBV perspective argue that, for a resource to confer competitive advantage to a firm, it must not be possessed by all competing firms, it must be difficult to imitate or duplicate through other means, and it must contribute positively to performance (Barney, 1991). Additionally, RBV researchers distinguish between resources that can be acquired in factor markets and those that are developed inside a firm. Resources and their performance effects have been conceptualized very broadly in previous research (Hoopes et al., 2003). For example, Tanriverdi and Venkatraman (2005) conceptualized various sources of knowledge as a firm-specific resource. Rigby and Zook (2002) show that the ability to combine internal and external information is a critical source of competitive advantage in their sample of innovative companies.

While individual routines used to collaborate with suppliers, customers, and internal work-teams contribute positively to firm performance, they are not idiosyncratic and cannot be considered a valuable resource in the RBV sense. In contrast, bundles of routines are more difficult to copy and less likely to be imitated by competitors or purchased in factor markets; they can be considered a valuable resource (Shah and Ward, 2003). The economic theory of complementarities informs us about the *super-additive* value of resource combinations (Cassiman and Veugelers, 2006). A set of resources is defined as complementary when using more of any one of the resources increases the returns from using others (Milgrom and Roberts, 1995). Conceptually, complementarity exists when a resource is more valuable in the presence of another resource than if it is considered alone (see Milgrom and Roberts, 1995 for a detailed mathematical treatment). In the context of this study, we argue that the joint involvement of suppliers, customers, and cross-functional internal employee teams in developing a new product leads to the development of super-additive synergies and is more valuable than the involvement of each individually. We refer to these super-additive synergies as "collaborative competence" and examine its conceptualization with the following proposition:

Proposition. *Collaborative competence is a multidimensional, higher order construct comprised of supplier involvement, customer involvement, and cross-functional team involvement.*

3.1. Impact on project performance

With increasing pressure on project managers and R&D personnel to develop products better and faster, the need to collaborate effectively and simultaneously with key stakeholders in the product development process has gained significant importance. Sutton and Hargaddon (1996) emphasize that, as a group engages in spanning the functional and organizational boundaries, more divergent thinking and combinatorial possibilities are likely and lead to higher group innovativeness. More recently, Wong (2004) shows that in creating an innovative solution, group members that seek external knowledge from across group boundaries also need to look inward, actively engaging in mutual discussions and brain storming internally to consider how external knowledge can be combined with their internal resources and capabilities to create innovative solutions.

Bringing the three distinct "thought worlds" (Dougherty, 1992) of suppliers, customers, and cross-functional teams together onto the same organizational platform forces each of them to understand and appreciate each other's concerns and work toward a mutually agreed-upon solution. Collective involvement of these diverse groups helps in developing a common language and a shared syntax for representing differences and understanding the critical interdependencies at boundaries (Carlile, 2004). It also facilitates a process of transforming and synchronizing localized capabilities and strengths into jointly produced knowledge that transcends local interests, thereby enabling early resolution of problems during the development process. For example, involving a focal firm's suppliers and customers together in the development process is beneficial because it allows the suppliers to integrate specific customer requirements into a component design at earlier design stages. At the same time, manufacturing involvement in the development process is also necessary in order for manufacturing function to initiate changes in manufacturing process technology and align its technological abilities and constraints within the product specifications. This also ensures that the unique capabilities of manufacturing process technology are taken into consideration by suppliers and the design team when decisions are made regarding the complexity and variety of components within product families. As a result, project performance goals are likely to be met, because wasteful expenditures in both materials and time can be minimized, delays in the development process can be prevented, and the product's time to market can be reduced. These arguments lead us to propose the following hypothesis.

Hypothesis 1. *Collaborative competence is positively associated with project performance.*

3.2. Impact on market performance

Research on collaboration across business units of multi-business firms has shown that sharing product designs, subsystems, components, and manufacturing processes creates "asset amortization benefits" from economies of scope for a firm (Tanriverdi and Venkatraman, 2005; Markides and Williamson, 1994). As business units share and reuse existing product knowledge, they tend to reduce development, tooling, and manufacturing costs, speeding up NPD activities and allowing the firm to rapidly address new market opportunities (Meyer and Lehnerd, 1997). We believe that these arguments are applicable in the context of an NPD project. Competitors may be able to achieve similarities in product designs in competing products, acquire similar marketing and advertising skills, and, to a lesser extent, implement similar managerial policies governing the development process. Their ability to recognize and imitate complementarities originating from inter- and intra-organizational collaboration routines within another firm, however, is significantly limited. Successfully recognizing and imitating complementarities require systemic changes across multiple organizational dimensions. Additionally, implementing a single dimension without implementing others is not likely to produce the intended performance improvements (Porter, 1996); rather, it may even produce negative performance effects (Milgrom and Roberts, 1995). Thus, in an NPD project, collaborative competence is a source of competitive advantage that leads to higher market performance of the product.

The development of the Boeing 787 Dreamliner is an example of how collaborative competence can be a unique source of competitive advantage and lead to market success. According to a recent news release from The Boeing Company (2007), the 787 Dreamliner is currently the fastest selling commercial airplane in aviation history. Boeing leveraged the technical capabilities of more than 50 partners spread over 130 locations and working together over a span of 4 years to develop the 787 Dreamliner. While similar technical capabilities were undoubtedly available to its competitors, Boeing's ability to integrate its partners into a network facilitated rapid technical knowledge exchange (MacCormack and Forbath, 2008). Such complementarity among the various dimensions of collaborative practices acts as a source of sustainable competitive advantage, which should lead to superior market performance. Therefore, we propose the following hypothesis.

Hypothesis 2. *Collaborative competence is positively associated with market performance.*

3.3. Relationship between project performance and market performance

The relationship between project and market performance has long been recognized as an important, yet unresolved, question in NPD research (Krishnan and Ulrich, 2001). A firm is more likely to develop successful products that have a higher customer value when its developmental activities are well-executed (Brown and Eisenhardt, 1995). Past research provides some evidence linking escalating product costs to market failure of a new product (Schmidt and Calantone, 1998; Boulding et al., 1997). A product pricing strategy and its unit cost may determine both its short- and long-term profitability (Tatikonda and Montoya-Weiss, 2001; Mallick and Schroeder, 2005). Similarly, the technical performance of the product may be an important determinant of its market performance. Cooper and Kleinschmidt (1987) have shown that superior technical performance of a product in terms of its features and innovativeness is a key factor in differentiating new product winners from losers. Time to market has also been an important determinant of market outcomes (Shankar, 1999; Bowman and Gatignon, 1996). Our review of literature suggests that, while individual measures at the operational level have been correlated with market performance measures, few prior studies (with the exception of Tatikonda and Montoya-Weiss, 2001) have systematically examined the overall impact of a holistic project performance measure on market performance. We address this issue in our study and examine the following hypothesis.

Hypothesis 3. *Project performance is positively associated with market performance.*

4. Methods

We use confirmatory factor analysis (CFA) and structural equation modeling (SEM) to measure and test our conceptual framework using survey data. All descriptive analyses are conducted using SPSS (12.1), and CFA and SEM are conducted using AMOS (4.0).

4.1. Sample

The data used for empirical analysis was collected as part of the third round of the High Performance Manufacturing (HPM) project in 2005–2006. The HPM was a large-scale, multi-country, multi-industry research project conducted by a team of international researchers and designed to comprehensively assess a manufacturing plant's operation. For Round 3, six countries were selected for study: Finland, Sweden, Germany, Japan, Korea, and the United States. These countries were chosen to obtain a broad representation of the industrialized world. In keeping with the first and second rounds, automotive suppliers, electronics, and machinery were selected as the three industries, using a stratified sample consisting of traditional and high performance manufacturing plants (for comprehensive details on data collection during Rounds 1 and 2, please see Flynn et al., 1995; Sakakibara et al., 1997).

Because participation in the study necessitated extensive effort on part of the plant, an assigned team member solicited plant participation by calling the plant manager of each prospective plant to describe the study objective, explain the participation mechanism, and relay the benefits of participation. Plants that agreed to participate appointed a survey coordinator to serve as the liaison with the research team; in many cases this liaison was the plant manager. A packet containing 21 surveys was then mailed to the coordinator who distributed the questionnaires to the appropriate respondents, collected the completed questionnaires, and mailed the responses back to the research team (for details on protocol followed in collecting data during Round 3, please refer to Peng et al., 2007).

The NPD survey instrument was new to Round 3 and was specifically developed to better understand the product development process. NPD researchers assisted in developing the survey instrument based on an extensive review of relevant literature. The instrument was pretested with academic and industry experts and at several manufacturing plants. During administration, each plant coordinator selected a product development manager as the key respondent for the NPD survey. To get accurate responses, the product development manager was asked to refer to a "recent major product development project" in which he/she participated, thus making a product development project the unit of analysis in this study. Our current study uses responses from the product development manager only. The final data set consists of 189 responses from product development managers relating one project to one plant. The response rate amounts to approximately 65 percent. Even so, we assessed nonresponse bias, comparing the plant size and annual revenue of the responding and non-responding plants. We did not find any evidence of a systematic non-response bias. Table 2 shows the profile of the sample across the different industries and countries.

Table 2
Profile of sample across different industries and countries.

Industry	Number of plants			
Country	Electronics	Machinery	Automotive	Total
Finland	14	6	10	30
Germany	9	13	19	41
Japan	10	11	13	34
South Korea	10	10	11	31
Sweden	7	10	7	24
United States	9	11	9	29
Total	59	61	69	189

4.2. Measures

4.2.1. Collaborative practices

We used 14 items on a 7-point Likert-type scale (1 = strong disagreement, 7 = strong agreement) to measure supplier involvement, customer involvement, and cross-functional involvement in the NPD process (see Appendix A). The scale items measure both the extent and the timing of the involvement of each group and are developed using the existing literature on collaboration (Brown and Eisenhardt, 1995; Ittner and Larcker, 1997; Hartley et al., 1997; Swink, 1999; Gerwin and Barrowman, 2002). For instance, the measurement items SUPINV2 (*We partnered with suppliers for the design of this product*), CUSTINV3 (*Customers were an integral part of the design effort for this project*), and CRFINV1 (*New product design teams have frequent interaction with the manufacturing function*) are each a measure of the extent to which the focal firm collaborates with suppliers, customers, and cross-functional teams in the product development process. Similarly, SUPINV1 (*Suppliers were involved early in the design efforts in this project*), CUSTINV1 (*We consulted customers early in the design efforts in this product*), and CRFINV2 (*Manufacturing is involved in the early stages of new product development*) are each a measure of the timing of involvement with suppliers, customers, and cross-functional teams.

4.2.2. Collaborative competence

A key tenet of complementarity theory is that the returns obtained from jointly using collaborative practices are greater than the sum of returns obtained from using individual practices in isolation (Milgrom and Roberts, 1995, p. 181). However, there are many different ways to model the joint adoption including a mathematical specification as illustrated by Milgrom and Roberts (1995) and statistical specification using interaction and latent model approaches. We model collaborative competence as a *reflective second order construct* to capture complementarities arising from the first order constructs. This second order, collaborative competence construct accounts for multi-lateral interactions and covariances among the first order constructs and is a superior statistical specification compared to pair-wise and higher order interactions. The use of second order construct to represent complementarities among first order constructs has been well documented in existing studies (Shah and Ward, 2007; Ettlie and Pavlou, 2006; Tanriverdi and Venkatraman, 2005; Sivadas and Dwyer, 2000). Using the second order construct also allows us to sidestep the statistical confounding that results from highly significant correlations among the first order constructs, which are consistent with the theory of complementarities (Whittington et al., 1999; Milgrom and Roberts, 1995).

4.2.3. NPD performance measures

Project performance is assessed using four commonly used items reflecting time-to-market, technical performance, unit manufacturing cost, and R&D budget as measured relative to goals (see Appendix A). These measures have been widely recognized as central objectives in measuring internal performance of an NPD process (Hauptman and Hirji, 1996; Wheelwright and Clark, 1992). Typically, these objectives are set at the start of project execution, and performance is assessed by comparing the objectives to the degree of achievement upon project completion. Thus, we asked NPD project managers to rate the success of the product development project relative to the set goals and objectives on the four measures. Market performance is measured using items for market share, overall profitability, return on investment, and overall commercial success (see Appendix A). These market performance measures have been frequently used in previous product development literature (Atuahene-Gima and Li, 2004; Brown and Eisenhardt, 1995). All responses concerning questions on project performance and market performance were measured on a 7-point Likert-type scale.

4.2.4. Control variables

Plant size is frequently associated with resource availability and bureaucratic structures. Larger plants have more resources to carry out product development tasks, but are also more likely to have longer development periods because of complex, formal structures (Ettlie, 1995). We measure plant size with the total number of hourly and salaried employees and use the logarithm of plant size to control its effects on project and market performance.

Another source of variation among product development projects comes from project team member characteristics, because team characteristics such as average experience of the team members play an important role in determining project outcomes (Ettlie, 1995). Teams with a concentration of specialists or members with relevant experience have a higher probability of achieving their project performance goals. We control for the effect of team characteristics on project performance and measure it using average team experience of the project team in years. Previous researchers have also noted the deleterious effects of large size on firm performance (Chandler, 1962; Child, 1972) because more employees are typically associated with increased bureaucracy and structural inertial forces (Hannan and Freeman, 1984) which negatively affect firm performance. Similar negative associations are also found in the context of team size and project performance. Thus, we include team size as a control variable on project performance.

Additionally, we control for the effects of product size (measured as the number of components in a complete product) on project performance. Products with a higher number of components are likely to have a higher degree of task interdependence than products with smaller numbers of components; this, in turn, may impact project performance outcomes.

Finally, previous NPD research has noted the impact of industry membership and country location on a firm's approach to product development (Ettlie, 1995; Hoskisson and Hitt, 1990). Therefore, in order to control for any heterogeneity among projects in the sample due to differences in their industry membership or their country location, we standardize the data across the six countries and three industries in our sample.

5. Results

We followed a two-stage procedure recommended by Anderson and Gerbing (1988) in testing our conceptual framework. During the first stage, we used CFA to develop and test the measurement models associated with collaborative practices and performance measures. This was done to purify and validate the constructs and to ensure that any misfit in the overall structural model representing our conceptual framework could be attributed to structural relationships. During the second stage, we use SEM to examine the hypothesized structural relationships. The model fit was assessed with goodness-of-fit index (GFI), incremental fit index (IFI), comparative fit index (CFI), Tucker Lewis index (TLI), and normed chi-square. We also report root mean square error of approximation (RMSEA) and its 90% confidence interval, root mean squared residual (RMSR), and Parsimony normed fit index (PNFI). Descriptive statistics and correlations among the variables are summarized in Table 3. The process details are described below.

5.1. Measurement model results

5.1.1. Collaborative practices

We analyzed four measurement models to establish the dimensional structure of collaborative practices using CFA.

Table 3
Means, standard deviations, and correlations of all variables[a].

	Variables	Mean	S.D.	1	2	3	4	5	6	7	8	9
1	Supplier involvement	4.63	1.14	1.00								
2	Customer involvement	4.92	1.14	0.42	1.00							
3	Cross-functional involvement	5.23	0.88	0.23	0.19	1.00						
4	Project performance	4.47	0.81	0.21	0.12	0.19	1.00					
5	Market performance	4.76	0.94	0.19	0.12	0.12	0.59	1.00				
6	Average team experience	11.14	11.18	0.08	0.00	0.13	0.04	0.00	1.00			
7	Team size	60.42	229.46	0.12	0.05	0.10	−0.04	−0.03	−0.02	1.00		
8	Product size	1721.70	4514.40	−0.08	−0.04	−0.03	0.08	0.17	0.01	0.03	1.00	
9	Plant size	1045.26	3145.52	−0.09	−0.15	−0.16	−0.01	−0.18	−0.02	0.10	0.02	1.00

[a] $n = 189$ observations; p-value < 0.05 for correlation values greater than 0.14; p-value < 0.01 for correlation values greater than 0.19.

Table 4a
Measurement models.

Models	χ^2 (df)	Normed χ^2	GFI	IFI	TLI	CFI	RMSEA (90% CI)	SRMR
Collaborative practices								
Model 1 (one-factor model)	415.73 (54)	7.70	0.70	0.55	0.45	0.55	0.19 (0.17, 0.21)	0.14
Model 2 (3 uncorrelated factors)	157.50 (54)	2.92	0.89	0.87	0.84	0.87	0.10 (0.08, 0.12)	0.16
Model 3 (3 correlated factors)	114.95 (51)	2.25	0.91	0.92	0.90	0.92	0.08 (0.06, 0.10)	0.06
Model 4 (one second order factor)	114.95 (51)	2.25	0.91	0.92	0.90	0.92	0.08 (0.06, 0.10)	0.06
Performance measures								
Model 5 (one-factor model)	56.57 (14)	4.04	0.91	0.90	0.84	0.90	0.13 (0.09, 0.16)	0.08
Model 6 (two-factor model)	14.87 (13)	1.14	0.98	0.99	0.99	0.99	0.03 (0.00, 0.08)	0.03

For more information on commonly recommended cutoff criterion for fit statistics, please refer to Shah and Goldstein (2006). Model 1: CFA involving all items representing the various collaborative practices loaded on one factor; Model 2: CFA involving the three uncorrelated first order factors; Model 3: CFA involving the three correlated first order factors; Model 4: CFA involving second order collaborative competence factor; Model 5: CFA involving all performance items on one factor; Model 6: CFA involving two correlated performance factors.

The fit statistics for these four models are shown in Table 4a. Model 1 conceptualizes that all 14 items representing the various collaborative practices are denoted by a unidimensional factor—i.e., the variance among all 14 items is accounted for by a single representative construct (also known as Harman's one-factor test). Not surprisingly, Model 1 has a very poor fit. In Model 2, we conceptualize that the 14 items are represented by three distinct and un-correlated first order factors corresponding to supplier involvement, customer involvement, and cross-functional involvement. The fit measures for Model 2 are significantly better than Model 1, indicating that a multi-dimensional model composed of three un-correlated first order factors is justified and is superior to a unidimensional first order factor model. Further evidence in support of Model 2 is obtained by examining the significance of the chi-square difference across the two models with respect to their degrees of freedom difference. The chi-square difference ($\Delta\chi^2 = 261.00$) is significant (p-value < 0.01). Two items (SUPINV5 and CUSINV3) in Model 2 had high inter-item correlations with other items and did not seem to add any incremental information to the measurement model. We dropped these two items from all further analysis (see Appendix A).

Model 3 conceptualizes that the three first order factors are free to correlate with each other (Fig. 1). The fit indices for Model 3 exceed the critical cutoff levels commonly used in empirical research (Shah and Goldstein, 2006). The superior fit is also reflected in the positive and significant correlations among the first order factors that represent collaborative practices. A comparison between Models 2 and 3 indicates that Model 3 represents the data considerably better than Model 2; the chi-square difference between the two models is also significant ($\Delta\chi^2 = 43.55$, $\Delta df = 3$, p-value < 0.01). Finally, in Model 4, we test the effectiveness of using a reflective second order construct to represent complementarities among the

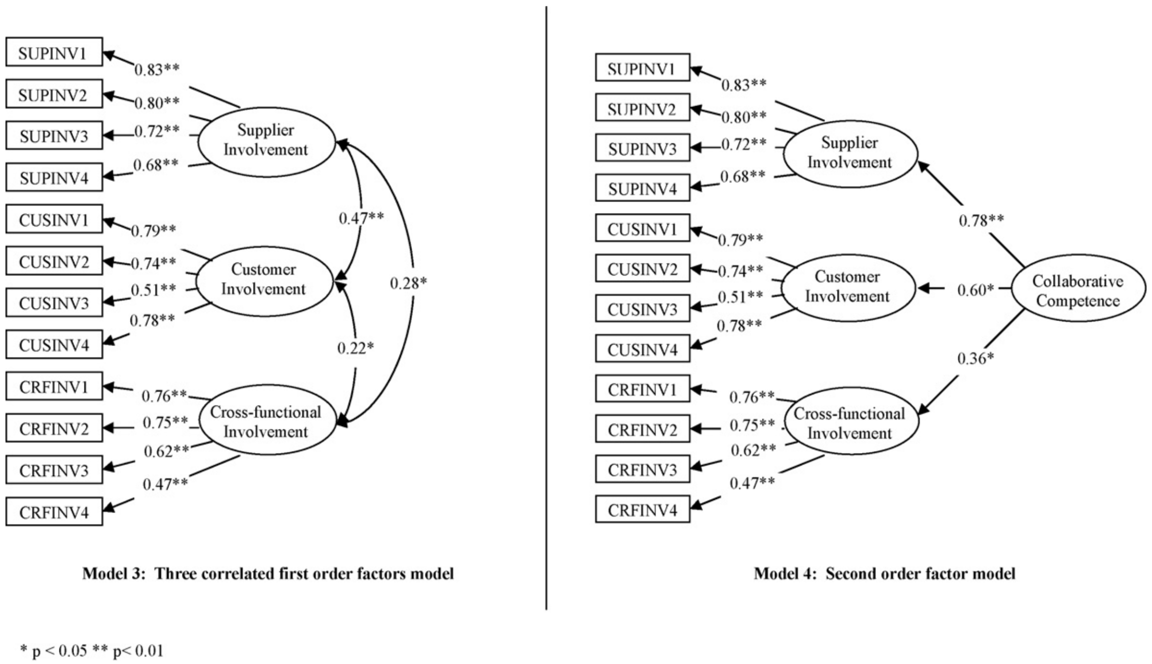

Fig. 1. Measurement models corresponding to Models 3 and 4 in Table 4a.

three first order factors (Fig. 1). When two nested models have exactly the same chi-square and fit measures (as do Models 3 and 4), the superiority of one model is established by examining the significance of the second order factor loadings in the measurement model on one hand and significance of the structural links in the second order factors model compared to the structural links in the first order factors model on the other (Venkatraman, 1989). All the second order factor loadings in Model 4 are significant (p-value < 0.05). Further, the structural model using the second order factor structure for collaborative competence fits the data better relative to the structural model with first order factors (discussed in Section 5.2). Both results strongly support the notion of complementarity and indicate that the second order factor structure is a better statistical specification for modeling collaborative competence.

Given the results, we used Model 4 to assess validity and reliability. We assessed convergent validity using standardized parameter loadings of the measurement items on their respective constructs. The standardized parameter loadings range from 0.51 to 0.83 and are significant at p-value < 0.01, providing a strong support of convergent validity. Discriminant validity is examined by calculating confidence intervals around the estimates of inter-factor correlation (Φ) (Anderson and Gerbing, 1988). Because none of the confidence intervals for the inter-construct correlations contain 1.0, we conclude that the constructs are distinct from each other. We assessed composite reliability using the procedures outlined by Fornell and Larcker (1981); the composite reliability values for each of the factors exceeds 0.70 (see Appendix A). As a set, the results indicate the dimensionality, validity, and reliability of collaborative competence as a second order factor and provide support to our proposition.

5.1.2. Performance measures

The dimensional structure for performance measures was assessed in a manner similar to measures for collaborative practices. We used CFA to analyze two measurement models. Model 5 consists of nine measures forming a unidimensional factor, whereas Model 6 consists of two distinct yet correlated factors representing project performance and market performance. Table 4a presents the fit statistics for these two models. We retain Model 6 because of its superior overall fit to the data; its fit indices dominate the fit statistics associated with Model 5 and exceed critical cutoff values. We also computed the chi-square difference across the two models and found it to be significant with respect to the corresponding change in the degrees of freedom ($\Delta\chi^2 = 38.47$, $\Delta df = 1$, p-value < 0.01). We dropped two items because they showed significant cross-loadings with other items.

Standardized parameter loading for all performance items on their respective constructs were significant and ranged from 0.58 to 0.84, indicating evidence of convergent validity. The confidence interval around the estimate of inter-factor correlation did not contain 1.0, indicating that project and market performance constructs are distinct from each other. The composite reliability for project performance and market performance were 0.66 and 0.80, respectively.

5.2. Structural model

The complete structural model representing our conceptual framework along with the parameter estimates is shown in Fig. 2. It includes the second order collaborative competence construct, and the three structural links representing our three hypotheses. We label this the complementarity model (Model 7). We also evaluate an alternate model shown in Fig. 3, where collaborative competence as the second order factor is removed; instead it includes the direct relationships between the three sets of collaborative practices and the two performance measures, along with the link from project to market performance. We label this the direct effects model (Model 8). Specifically, the complementarity model, with its second order factor, is more parsimonious because it requires fewer parameters to be estimated, compared to the direct effects model, and accounts for the covariance among the first order factors. We discuss the model statistics and their implications below.

In comparing the complementarity model with the direct effects model, we find that the fit statistics for the complementarity model are marginally better than those of the direct effects model (Table 4b). More importantly, only two of the six direct structural links from cross-functional involvement, customer involvement, and supplier involvement to project and market performance are significant (at p-value < 0.05) in the direct effects model, indicating poor model specification. The insignificant structural links are inconsistent with the existing literature and provide additional indirect support to the complementarity model. We also compare the variance explained by the complementarity and direct effects models. Because the link between project performance and market performance is similar in algebraic sign, magnitude, and significance in the two models, we assess the variation (R^2) explained in the project performance by the two models; the complementarity model explains 34% ($R^2 = 0.34$) variation compared to the 26% ($R^2 = 0.26$) of the variation explained by the direct effects model. This clearly demonstrates the superiority of the complementarity model over the direct effects model in explaining the variation in project performance.

Consequently, we use the complementarity model to evaluate Hypotheses 1, 2, and 3. We use the significance of the structural links represented by standardized parameter

Table 4b
Structural models.

Models	χ^2 (df)	Normed χ^2	GFI	IFI	TLI	CFI	RMSEA (90% CI)	SRMR	PNFI
Model 7 (Complementarity model)	349.63 (223)	1.57	0.87	0.90	0.88	0.90	0.05 (0.04, 0.07)	0.07	0.67
Model 8 (Direct effects model)	386.23 (222)	1.74	0.86	0.87	0.85	0.87	0.06 (0.05, 0.07)	0.10	0.65

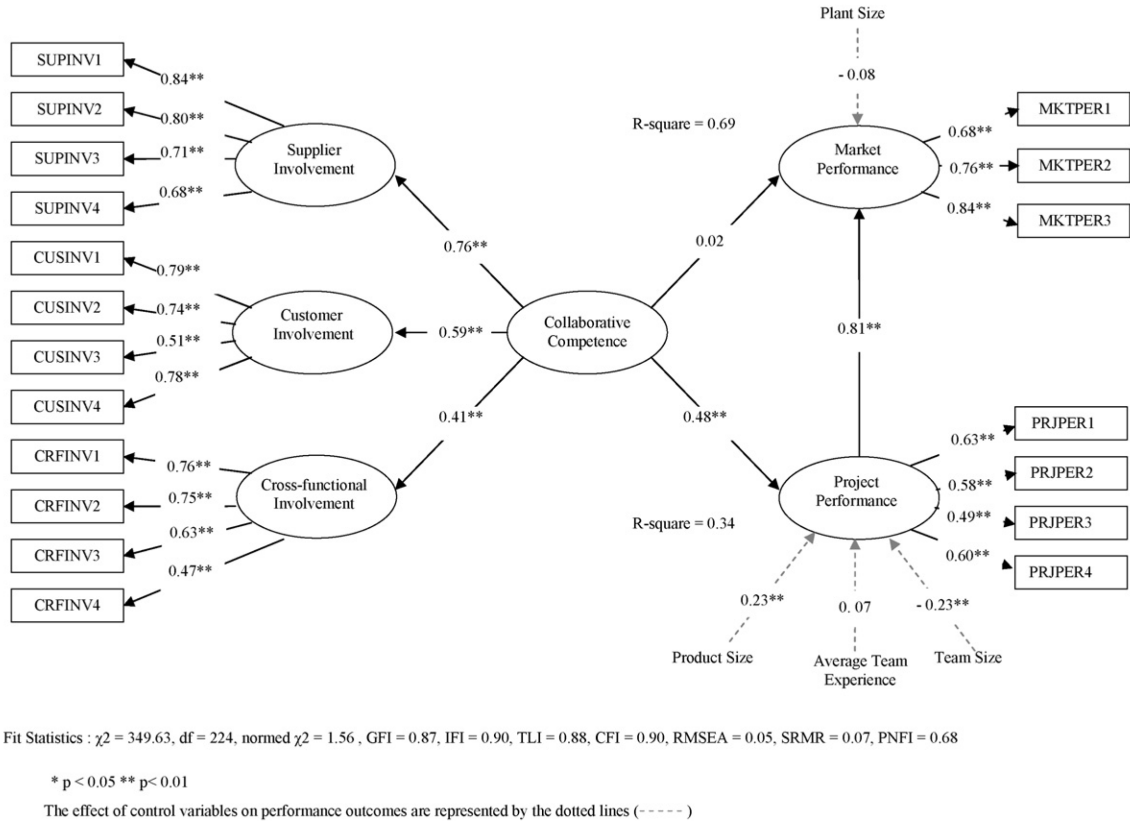

Fig. 2. Complete structural model representing the complementarity model (Model 7).

estimates associated with the structural paths in evaluating the hypotheses. As indicated in Fig. 2, the link from collaborative competence to project performance is positive and significant (structural link = 0.48, p-value < 0.01), providing strong support for Hypothesis 1. In contrast, the path from collaborative competence to market performance is not significant (at p-value < 0.05); therefore, we do not find support for Hypothesis 2. Hypothesis 3 proposed a positive impact of project performance on market performance; our results provide significant support for that relationship (structural link = 0.81, p-value < 0.01).

As expected, among the control variables in the complementarity model, project team size has a significant negative effect on project performance, indicating that larger teams are associated with lower project performance. In contrast, the effect of product size on project performance is significant and positive, suggesting that a product with a larger number of components is associated with higher level of project performance. This result is counter-intuitive and in contradiction with previous research. Perhaps the relationship between product size and project performance is contingent on the number of new components designed exclusively for the product or on component commonality among different product configurations, so the larger product size does not necessarily imply greater task interdependence or complexity of product development efforts. Future research should examine both the direct relationship between product size and project performance and the contingency effects of other variables impacting their direct relationship.

6. Discussion and conclusion

The goal of this study was to investigate the performance benefits of simultaneously involving suppliers, customers, and cross-functional teams in the NPD process. We use the resource-based view and the complementarity theory to conceptualize synergies that arise from involving suppliers, customers, and cross-functional internal teams in the NPD process as a second order factor and label it "collaborative competence." Using multi-country, multi-industry, project level data from 189 manufacturing plants, we empirically validate the second order factor structure. We then assess the performance effects of collaborative competence on project and market-based performance. We discuss theoretical, empirical, and managerial contributions below.

Theoretically, this study overcomes weaknesses observed in prior NPD research. Most previous NPD studies focus on limited aspects of involvement and examine sources of synergy obtained from these domains in an isolated manner. Such fragmented conceptualization does not represent the product development process in practice because it ignores the "interdependencies" (Krishnan and Ulrich, 2001) and the "shared product concept" (Leonard-Barton, 1995) that exist among suppliers, customers, and cross-functional teams. To better

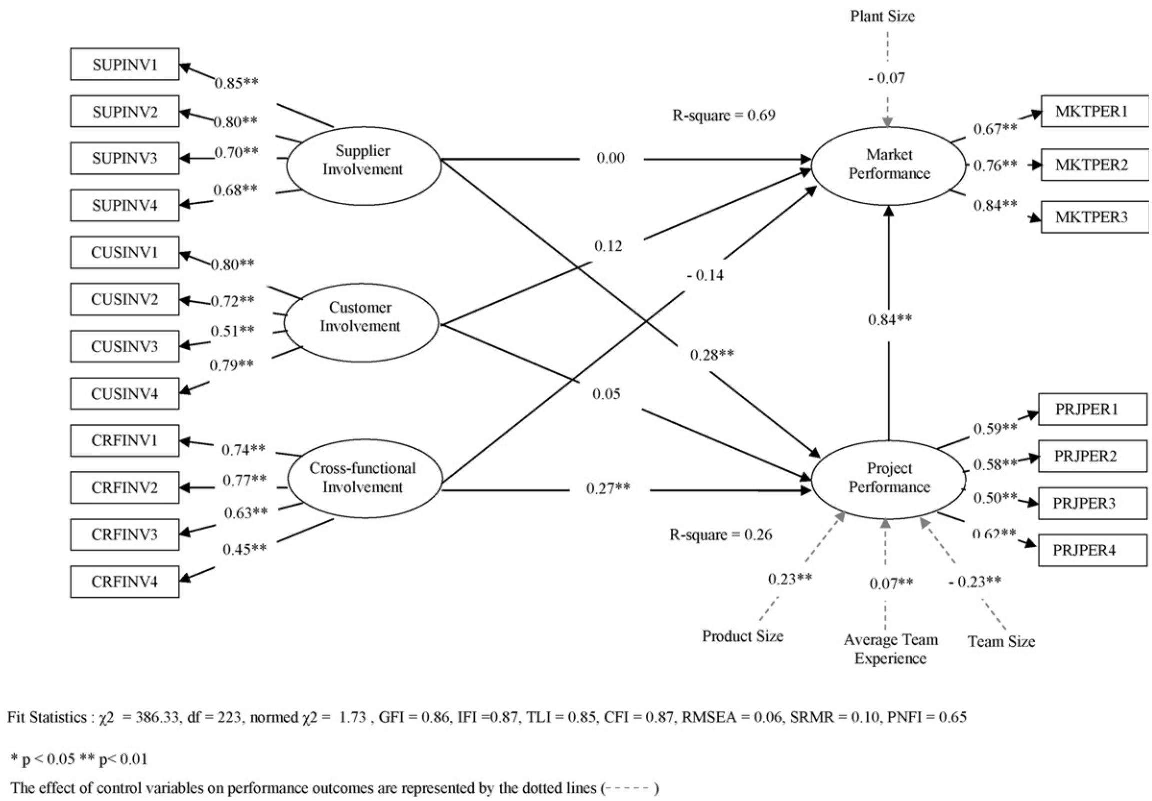

Fig. 3. Complete structural model representing the direct effects model (Model 8).

represent these interdependencies, our study conceptualized collaborative competence as the synergy resulting from simultaneous involvement of the three key groups associated with the NPD process. Further, while the resource-based view recognizes that the separate involvement of supplier, customer, or cross-functional teams are each valuable and create supplier involvement synergies, customer involvement synergies, and cross-functional involvement synergies respectively, it does not recognize the super-additive synergies arising from involving all three groups (suppliers, customers, and cross-functional teams) in the product development process. Our use of the complementarity theory in this study is an effort toward presenting a more nuanced and realistic picture of collaborative product development.

With its second order, multi-dimensional conceptualization of collaborative competence, this study captures both the synergies arising from the independent involvement of supplier, customer, and cross-functional teams and the super-additive value of synergies arising from the complementarity of different types of involvement. As the rich literature stream on the relationship between inter- and intra-organizational involvement and firm performance indicates, conceptualizing individual dimensions of a multi-dimensional construct as independent and examining their performance effects separately may lead to inconsistent and ambiguous results.

We make three empirical contributions to the existing literature. First, we establish that the collective pursuit of collaborative practices such as supplier involvement, customer involvement, and cross-functional involvement underlies the higher order collaborative competence. Although the usefulness of each of the three collaborative practices has been cited in previous literature (Gerwin and Barrowman, 2002; Krishnan and Ulrich, 2001), they have not been modeled together. The superiority of the second order factor model (Model 4) compared to the three correlated factors model (Model 3) confirms that multiple manifestations of involvement are all driven by a cohesive, yet unobservable, underlying competence. Further, our analysis shows that the complementarity model (Model 7), compared to the direct effects model (Model 8), yields statistically stronger and practically valid results. Specifically, the extent of variation in project performance explained by the second order latent construct is higher than that produced by separately analyzing the three sets of collaborative practices. Overall, our efforts to present a parsimonious representation of interdependencies among collaborative practices using a second order latent construct is both conceptually and methodologically a step toward providing a nuanced perspective on the integrated nature of the product development process.

Our second contribution consists in demonstrating that collaborative competence impacts project and market performance differently. While collaborative competence has a positive and significant impact on project performance, its impact on market performance is insignificant. We also find a significant positive link between project and market performance. As a set, the results associated with H2 and H3 suggest that the link between collaborative competence and market performance is *indirect*, mediated through project performance. We use the Sobel test to examine the significance of the mediation effect, an approach advocated by Baron and Kenny (1986). The Sobel test statistic equals 2.83 and is significant (p-value < 0.01). The analyses show that the link between collaborative competence and market performance becomes insignificant when project performance is added to the model, suggesting a complete mediation relationship between collaborative competence and market performance (Venkatraman, 1989). An insignificant direct link between collaborative competence and market performance is surprising given that customer involvement in NPD projects forms a key component of this competence. The primary objective that drives firms to involve customers in their product development efforts is to understand customer requirements and market trends so that superior market performance can be achieved.

What, then, explains our finding? We believe that there are two potential explanations for the insignificant direct link between collaborative competence and market performance. One is that project execution holds the key to market success and that achieving superior market performance from inter- and intra-organizational involvement necessitates excellence on project performance metrics. Specifically, companies that fail to achieve their desired project performance outcomes will also fail in achieving market performance goals. This interpretation puts significant pressure on firms to excel on operational level metrics as a precondition to excelling in the market. In general, this pattern of results, where an operational level variable is found to have an impact on operational level performance but not on market or financial measures of performance, is similar to that observed in supply chain and information technology literature. For instance, Vickery et al. (2003, p. 535) found that coordination of value-added activities has a direct impact on competitive performance, but not on the financial performance of the firm. Similarly, information technology researchers have called the general failure of studies to show a direct positive relationship between information technology investment and measures of overall financial performance profitability a paradox (Dedrick et al., 2003, p. 23). Barua et al. (1995) argue that the impact of operational variables on overall firm performance should be easier to detect through intermediate measures of operational performance. A second explanation for the insignificant effects of collaborative competence on market performance could be that perhaps the collaboration items used in our study (particularly, the customer involvement scale) lack sufficient variation to capture the true impact of collaborative competence on market performance. This is plausible, given that the collaboration scales used in our study have been derived primarily from existing studies, where they have gradually evolved and been optimized toward measuring operational performance benefits rather than market or financial performance metrics.

Our third contribution derives from the increased generalizability of our findings due to the broad scope of the sample. Our multi-industry, multi-country sample stands in sharp contrast to many empirical studies in the NPD literature wherein the sample of companies used is simply either a set of Japanese companies or U.S. companies based in the electronics or the computer industry (Balachandra and Friar, 1997). As such, the generalizability of previous empirical findings is primarily restricted to high performing firms in a single highly competitive industry and does not portray the true state of affairs across the whole landscape of product development efforts in various industries and countries.

Although the conceptual framework in our study examines the performance impact of simultaneously involving suppliers, customer, and cross-functional teams in the NPD process, its implications are not restricted only to projects requiring both inter- and intra-organizational collaboration. For example, NPD projects that are completely internal to a firm may involve multiple stakeholders (e.g., in-house IT projects which involve coordination between the internal software team and multiple departments). In these cases, harnessing the knowledge complementarities across multiple stakeholders by jointly involving them in the project is likely to lead to enhanced performance outcomes.

Our study also has important managerial implications. While managers must continue to involve a diverse group of stakeholders in their product development process to achieve superior performance, the real operational benefits of collaboration are derived when efforts are made to synchronize localized capabilities and strengths into jointly produced knowledge that transcends local interests and enables early resolution of problems during the NPD process. Further, achieving superior market performance impact from competence in involving these diverse groups is contingent on how the competence affects intermediate operational level performance. Thus, product development managers must continue to focus on ensuring that efficiency at the operational level is achieved, because it constitutes the first step toward improvement in overall firm performance.

These managerial implications can also be extended beyond the NPD process and into any context that involves alliances or firms engaging in an outsourcing/offshoring relationship with a vendor firm. While potential savings from economies of scope or labor arbitrage are certainly a key reason for firms to engage in such relationships, it is unlikely that such benefits will accrue unless operational performance first improves. As a recent AT Kearney (2007) study on offshoring indicates, that the best performing companies in their survey "focus less on saving money and more on improving operational performance, and in doing so they save more money" (p. 11). That is, the surveyed companies that focused on improving operational metrics achieved the largest cost savings.

7. Future research and limitations

While our study highlights the advantages of collaborating simultaneously with suppliers, customers, and cross-functional teams in the NPD process, firms may not use the three sets of collaborative practices to an equal extent.

Specifically, a focal firm's choice of collaborating with suppliers or customers may depend on contextual and situational conditions on one hand and the supply chain structure on the other. Future research efforts directed at studying the collaboration process in NPD would greatly benefit by examining the conditions under which an unbalanced collaborative approach may result in superior outcomes compared to a more balanced collaborative approach.

Another potential area for future research would be to examine the variation in the costs of collaboration across the three interfaces. At a macro level, collaborative practices with suppliers, customers, and cross-functional teams are similar in that they involve the sharing of technical and business knowledge. However, collaboration costs across each of the three interfaces could vary significantly due to the differences in their level of "shared context" (Hinds and Mortensen, 2005)—arising from commonality in information, tools, work processes, and work cultures across the three interfaces—with the core design team. Because cross-functional teams are a part of the same focal firm as the core design team, their levels of shared context with the core design team are likely to be higher compared to that of suppliers and customers. Consequently, there are fewer barriers for information flow between the cross-functional teams and the core design team, and the potential for conflict is low.

Additionally our study is conducted from the point of view of the focal firm; focusing on a focal firm inherently misses the supplier–customer aspect embodied by the supply chain perspective. Suppliers and customers are each likely to have an independent and different interpretation of the term 'collaboration', and evaluate the extent of its collaboration efforts with the focal firm differently. A more nuanced understanding of collaboration and the effectiveness of collaborative competence on performance outcomes require focusing on a triad (Supplier-Focal Firm-Customer) as the unit of analysis. We call upon future researchers to extend examining collaboration from a dyadic to a triadic perspective.

Future research might also benefit from developing measurement items that capture aspects of involvement other than the extent and timing of involvement. While our measures demonstrate interesting results, more finely grained measurement items will help capture greater variation across the three sets of collaborative practices. New measurement items may enable researchers to use other research methods in examining the performance impact of collaborative competence. For instance, using categorical variables to measure collaboration allows researchers to use Milgrom and Roberts' (1995) approach. Specifically, one can compare the project performance of firms that use all three sets of collaborative practices to the project performance of firms that use any two sets or even only one set. Cassiman and Veugelers (2006) provide an excellent example of this approach. Combining analytical and empirical methods such as agent-based simulation using NK model seems a promising approach for testing complementarity; Porter and Siggelkow (2008) provide a detailed description of this approach.

As with any study, our results must also be considered in light of the nature of the research sample and data. We collected data from a single key respondent; while this is currently the standard method in empirical research, it is associated with common method bias. To assess common method bias, we looked for a general methods factor during CFA, but did not find one. We also attempted to minimize its impact through careful selection of the respondents—for this study we used responses only from NPD managers. Although the relationships we examined are derived from theory, any interpretation of causality between constructs should be treated with caution. Another limitation of our study is the absence of control variables related to the suppliers' and customers' involvement. Subsequent research could replicate the results obtained from this study by including control variables such as number of suppliers/customers, previous experience in working with the suppliers/customers, duration and terms of the relationship with suppliers/customers, etc. Despite these limitations, our multi-country, multi-industry study provides researchers and managers with clear guidance for achieving superior project and market performance.

Appendix A. Measures

Responses to supplier involvement, customer involvement and internal involvement measures are given on 7-point Likert-type scale (1 = strong disagreement, 7 = strong agreement).

Supplier involvement (ρ = 0.84)	
SUPINV1	Suppliers were involved early in the design efforts, in this project.
SUPINV2	We partnered with suppliers for the design of this product.
SUPINV3	Suppliers were frequently consulted about the design of this product.
SUPINV4	Suppliers were an integral part of the design effort.
SUPINV5	Suppliers were selected after the design for this product was completed[a,b].
Customer involvement (ρ = 0.80)	
CUSTINV1	We consulted customers early in the design efforts for this product.
CUSTINV2	We partnered with customers for the design of this product.
CUSTINV3	Customers were frequently consulted about the design of this product[a].
CUSTINV4	Customers became involved in this project only after the design was completed[b].
CUSTINV5	Customers were an integral part of the design effort for this project.
Cross-functional involvement (ρ = 0.75)	
CRFINV1	New product design teams have frequent interaction with the manufacturing function.
CRFINV2	Manufacturing is involved in the early stages of new product development.
CRFINV3	The manufacturing function is involved in the creation of new product concepts.
CRFINV4	New product concepts are developed as a result of the involvement of various functions.

[a] Deleted items.
[b] Reverse coded items.

Appendix A (*Continued*)

Please rate the success of this product development project, relative to its goals (1 = significantly worse, 2 = worse, 3 = somewhat worse, 4 = about same, 5 = somewhat better, 6 = better, 7 = significantly better).

Market performance (ρ = 0.80)	
MKTPER1	Market share.
MKTPER2	Overall profitability.
MKTPER3	Return on investment.
MKTPER4	Overall commercial success[a].
Project performance (ρ = 0.66)	
PRJPER1	Technical performance relative to specifications.
PRJPER2	Time to market.
PRJPER3	Unit manufacturing cost.
PRJPER4	R&D budget.
Control variables	
Average team experience	Approximate average years of experience of team members.
Product size	How many parts are required to make one unit?
Team size	The total number of individuals involved in the project.
Plant size	Number of hourly personnel + number of salaried personnel.

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